Ritam Basu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	6.88
Vivekananda Mission School, Joka	Class XII (ISC)	2023	85.60%
Vivekananda Mission School, Joka	Class X (ICSE)	2021	90.60%

POSITIONS OF RESPONSIBILITY

Public Relations & Outreach Coordinator | Finance & Consulting Club, IIT Hyderabad

- **Scaled the club's social media following from 800 to 2,300+** by developing targeted content and event-led growth campaigns, driving stronger student reach and engagement.
- Owned outreach and execution of **speaker-led events**, successfully coordinating the engagement and session with **finance influencer Neha Nagar** from initial outreach through promotion and on-ground execution.
- Developed and executed marketing campaigns for **flagship events and competitions**, coordinating across teams to strengthen brand visibility and student participation.

Core Member | Elektronika (Technical Club), IIT Hyderabad

- Developed hands-on biomedical engineering projects including a real-time ECG-based heart rate monitor and an EMG controlled bionic robotic arm, applying embedded systems and signal processing skills.
- Contributed to club technical initiatives by building and documenting project workflows, and collaborated with peers to explore applications of biomedical sensors in assistive technology.

SKILLS

Management/Business: Event Management, Market Research, Marketing & Outreach, Financial Analysis, Consulting Frameworks, Team Management, Pitching, Public Speaking, Client Management, Product Sense, Project Management

Technical: C++, Python, MATLAB, Arduino, Excel, Kotlin, Android Studio

Collaboration & Communication: Stakeholder Engagement, Cross-functional Collaboration, Content Creation, Pitching & Presentation

RELEVANT COURSES

Relevant Coursework: Corporate Finance, Decision Modelling, Innovation Management, Economics for Technocrats, Finance for Engineers, Fundamentals of ESG for Entrepreneurs

The Complete Android 14 & Kotlin Development Masterclass - Udemy

Completed a comprehensive 66.5-hour course covering Android 14 app development using Kotlin, building functional applications with modern Android architecture components.

Gained hands-on experience in UI/UX design with Jetpack Compose, API integration, and Kotlin-based programming for mobile development.

PROJECTS

meRead | Personal Project | Kotlin, Jetpack Compose, Android Studio

- Developed a **Duolingo-inspired reading and learning app** as a personal project, designed to make reading more engaging through gamified challenges, streaks, progress tracking, and interactive content.
- Designed and implemented the **Android UI and user experience using Jetpack Compose**, building a structured reading flow focused on consistency, engagement, and measurable user progress.

Justdial Clone | Front-End MVP | Next.js, React, Tailwind CSS

- Developed the **front-end MVP of a Justdial-inspired local commerce platform**, building a map-based interface for users to discover nearby businesses, products, and services.
- Designed responsive UI flows for **location-based search, listings, product discovery, and seller exploration**, translating the core Justdial experience into an interactive web prototype.

Heart Rate Monitor using ECG

Developed a real-time heart rate monitoring system using the ECG Pill sensor from Upside Down Labs interfaced with Arduino Uno, enabling live physiological data acquisition.

Implemented a peak detection algorithm in MATLAB to accurately identify R-peaks in the ECG waveform and compute live heart rate in beats per minute. Demonstrated end-to-end pipeline from hardware signal acquisition to software-based signal processing, bridging biomedical instrumentation with computation.

Laser-Based Security Alarm System | *Arduino Uno, C++, Embedded Systems*

- Built a laser-based intrusion detection system using an **LDR, laser module, relay, buzzer, and light-blocking sensor**.
- Programmed Arduino to detect laser beam interruptions, trigger a **10-second alarm**, and control an **AC bulb** via a relay.
- Implemented independent sensor logic for object detection and real-time hardware control, demonstrating embedded programming and sensor interfacing skills.

Bionic Robotic Arm using EMG | Arduino, MATLAB, Machine Learning

- **Developed a gesture-controlled bionic robotic arm** using the **EMG Pill sensor from Upside Down Labs** and Arduino, acquiring raw sEMG signals from individual finger movements to control servo-driven arm movements.
- Collected and labelled a **custom EMG dataset** for rest, thumb, index, and middle-finger gestures, extracted signal features, and trained a **machine learning classifier** for real-time gesture recognition and robotic arm actuation.

EXTRACURRICULAR

NSO Table Tennis Player, IIT Hyderabad *(2025-Present)*Represented the institute as part of the National Sports Organization (NSO) Table Tennis program, demonstrating teamwork, discipline, and consistent commitment to competitive sports.

Volunteer | International Relations Cell, IIT Hyderabad Interacted with Japanese and international delegates, coordinating city tours and providing logistical support while fostering a welcoming and culturally inclusive environment.